

Technical Maintenance & Equipment Inspection

Date: 2026-09-27

3. ISSUE DESCRIPTION & ROOT CAUSE

5. REPLACED / REQUIRED SPARE PARTS

6. SITE & FAULT PHOTOGRAPHIC EVIDENCE (12 Photos)

Page 1/3 - Confidential Technical Service Record

FIELD SERVICE REPORT

Technical Maintenance & Equipment Inspection

Service Training

6.2 POWER UNITS

GENERAL INFORMATION

Initial, the truck has three power units. Their design and operating principles are completely identical. The power unit 1A1 and 1A2 control the two traction motors. The power unit 2A1 controls the pump motor. The power units are equipped with a 20-pin connector through which they are connected to the control system. The power units are equipped with a 20-pin connector through which they are connected to the control system. The power units are equipped with a 20-pin connector through which they are connected to the control system. The power units are equipped with a 20-pin connector through which they are connected to the control system.

Each power unit controls its assigned motor independently. The power unit controls the complete control system for generation of the three phase current for the motor. Via the CANbus, the power unit receives information about the speed torque and the speed of the motor from the control system.

The power unit regulates the current feedback value (torque) which is determined by the motor current. Thermal sensors are provided in the area of the power transistor to monitor the temperature of the transistors. As soon as the transistor temperature reaches 85 °C, the motor current is reduced continuously. Monitoring of the temperature and the associated power reduction is complete shut down as indicated by the controller.

The capacitors in the intermediate circuit are located underneath the cover. The capacitors are required for buffering the battery voltage.

The complete control system for the transmission and the monitoring device is integrated in the power section. This means that the power unit 6.2.2 is not required in case of an error. It sends a corresponding signal to the control system via the CANbus.

The speed sensors and the temperature sensors of the motors are connected to the power units. The information about motor speed and temperature is sent to the control system via the CANbus. The speed sensors get a supply voltage of 15 V. This voltage is generated in the power unit.

Each power unit may be requested separately by the control system (enable signal). If the power unit works properly, it sends a feedback signal to the control system.

IMPORTANT NOTE
The power units may reverse speed, since this could lead to destruction of the existing hardware. Do not touch the connector pins, since this could result in destruction of internal components by electrostatic charge. The power units complete all control at the 20-pin connector are available through an spare parts department. The cover protects the connections from electrostatic charge. This cover must be installed whenever the power units are transported.

Service Training

6.2.2 POWER UNIT

GENERAL INFORMATION

Initial, the truck has three power units. Their design and operating principles are completely identical. The power unit 1A1 and 1A2 control the two traction motors. The power unit 2A1 controls the pump motor. The power units are equipped with a 20-pin connector through which they are connected to the control system. The power units are equipped with a 20-pin connector through which they are connected to the control system. The power units are equipped with a 20-pin connector through which they are connected to the control system. The power units are equipped with a 20-pin connector through which they are connected to the control system.

Each power unit controls its assigned motor independently. The power unit controls the complete control system for generation of the three phase current for the motor. Via the CANbus, the power unit receives information about the speed torque and the speed of the motor from the control system.

The power unit regulates the current feedback value (torque) which is determined by the motor current. Thermal sensors are provided in the area of the power transistor to monitor the temperature of the transistors. As soon as the transistor temperature reaches 85 °C, the motor current is reduced continuously. Monitoring of the temperature and the associated power reduction is complete shut down as indicated by the controller.

The capacitors in the intermediate circuit are located underneath the cover. The capacitors are required for buffering the battery voltage.

The complete control system for the transmission and the monitoring device is integrated in the power section. This means that the power unit 6.2.2 is not required in case of an error. It sends a corresponding signal to the control system via the CANbus.

The speed sensors and the temperature sensors of the motors are connected to the power units. The information about motor speed and temperature is sent to the control system via the CANbus. The speed sensors get a supply voltage of 15 V. This voltage is generated in the power unit.

Each power unit may be requested separately by the control system (enable signal). If the power unit works properly, it sends a feedback signal to the control system.

IMPORTANT NOTE
The power units may reverse speed, since this could lead to destruction of the existing hardware. Do not touch the connector pins, since this could result in destruction of internal components by electrostatic charge. The power units complete all control at the 20-pin connector are available through an spare parts department. The cover protects the connections from electrostatic charge. This cover must be installed whenever the power units are transported.

1 Negative connection of power unit
2 Positive connection of power unit
3 Phase 1 - motor connection
4 Phase 2 - motor connection
5 Phase 3 - motor connection
6 Phase 4 - motor connection



Service Training

Phase voltage feedback

Based on the information on the desired speed and torque, received via CANbus, each power unit generates the three phase voltage for its motor independently. The feedback signal of the individual phase currents and phase voltages of the motors are also sent via CANbus. For safety reasons, the individual phase voltages of the traction motors are additionally sent to the control system via a separate line.

Traction motor 1M1 voltage L1 X13.24

Traction motor 1M1 voltage L2 X13.43

Traction motor 1M1 voltage L3 X13.62

Traction motor 1M2 voltage L1 X13.61

Traction motor 1M2 voltage L2 X13.80

Traction motor 1M2 voltage L3 X13.81

Traction motor 1M3 voltage L1 X13.81

Traction motor 1M3 voltage L2 X13.81

Traction motor 1M3 voltage L3 X13.81

Traction motor 1M3 voltage L4 X13.81

Traction motor 1M3 voltage L5 X13.81

Traction motor 1M3 voltage L6 X13.81

Traction motor 1M3 voltage L7 X13.81

Traction motor 1M3 voltage L8 X13.81

Traction motor 1M3 voltage L9 X13.81

Traction motor 1M3 voltage L10 X13.81

Traction motor 1M3 voltage L11 X13.81

Traction motor 1M3 voltage L12 X13.81

Traction motor 1M3 voltage L13 X13.81

Traction motor 1M3 voltage L14 X13.81

Traction motor 1M3 voltage L15 X13.81

Traction motor 1M3 voltage L16 X13.81

Traction motor 1M3 voltage L17 X13.81

Traction motor 1M3 voltage L18 X13.81

Traction motor 1M3 voltage L19 X13.81

Traction motor 1M3 voltage L20 X13.81

Traction motor 1M3 voltage L21 X13.81

Traction motor 1M3 voltage L22 X13.81

Traction motor 1M3 voltage L23 X13.81

Traction motor 1M3 voltage L24 X13.81

Traction motor 1M3 voltage L25 X13.81

Traction motor 1M3 voltage L26 X13.81

Traction motor 1M3 voltage L27 X13.81

Traction motor 1M3 voltage L28 X13.81

Traction motor 1M3 voltage L29 X13.81

Traction motor 1M3 voltage L30 X13.81

Traction motor 1M3 voltage L31 X13.81

Traction motor 1M3 voltage L32 X13.81

Traction motor 1M3 voltage L33 X13.81

Traction motor 1M3 voltage L34 X13.81

Traction motor 1M3 voltage L35 X13.81

Traction motor 1M3 voltage L36 X13.81

Traction motor 1M3 voltage L37 X13.81

Traction motor 1M3 voltage L38 X13.81

Traction motor 1M3 voltage L39 X13.81

Traction motor 1M3 voltage L40 X13.81

Traction motor 1M3 voltage L41 X13.81

Traction motor 1M3 voltage L42 X13.81

Traction motor 1M3 voltage L43 X13.81

Traction motor 1M3 voltage L44 X13.81

Traction motor 1M3 voltage L45 X13.81

Traction motor 1M3 voltage L46 X13.81

Traction motor 1M3 voltage L47 X13.81

Traction motor 1M3 voltage L48 X13.81

Traction motor 1M3 voltage L49 X13.81

Traction motor 1M3 voltage L50 X13.81

Traction motor 1M3 voltage L51 X13.81

Traction motor 1M3 voltage L52 X13.81

Traction motor 1M3 voltage L53 X13.81

Traction motor 1M3 voltage L54 X13.81

Traction motor 1M3 voltage L55 X13.81

Traction motor 1M3 voltage L56 X13.81

Traction motor 1M3 voltage L57 X13.81

Traction motor 1M3 voltage L58 X13.81

Traction motor 1M3 voltage L59 X13.81

Traction motor 1M3 voltage L60 X13.81

Traction motor 1M3 voltage L61 X13.81

Traction motor 1M3 voltage L62 X13.81

Traction motor 1M3 voltage L63 X13.81

Traction motor 1M3 voltage L64 X13.81

Traction motor 1M3 voltage L65 X13.81

Traction motor 1M3 voltage L66 X13.81

Traction motor 1M3 voltage L67 X13.81

Traction motor 1M3 voltage L68 X13.81

Traction motor 1M3 voltage L69 X13.81

Traction motor 1M3 voltage L70 X13.81

Traction motor 1M3 voltage L71 X13.81

Traction motor 1M3 voltage L72 X13.81

Traction motor 1M3 voltage L73 X13.81

Traction motor 1M3 voltage L74 X13.81

Traction motor 1M3 voltage L75 X13.81

Traction motor 1M3 voltage L76 X13.81

Traction motor 1M3 voltage L77 X13.81

Traction motor 1M3 voltage L78 X13.81

Traction motor 1M3 voltage L79 X13.81

Traction motor 1M3 voltage L80 X13.81

Traction motor 1M3 voltage L81 X13.81

Traction motor 1M3 voltage L82 X13.81

Service Training

Temperature information 1

Temperature information 2

Temperature information 3

Temperature information 4

Temperature information 5

Temperature information 6

Temperature information 7

Temperature information 8

Temperature information 9

Temperature information 10

Temperature information 11

Temperature information 12

Temperature information 13

Temperature information 14

Temperature information 15

Temperature information 16

Temperature information 17

Temperature information 18

Temperature information 19

Temperature information 20

Temperature information 21

Temperature information 22

Temperature information 23

Temperature information 24

Temperature information 25

Temperature information 26

Temperature information 27

Temperature information 28

Temperature information 29

Temperature information 30

Temperature information 31

Temperature information 32

Temperature information 33

Temperature information 34

Temperature information 35

Temperature information 36

Temperature information 37

Temperature information 38

Temperature information 39

Temperature information 40

Temperature information 41

Temperature information 42

Temperature information 43

Temperature information 44

Temperature information 45

Temperature information 46

Temperature information 47

Temperature information 48

Temperature information 49

Temperature information 50

Temperature information 51

Temperature information 52

Temperature information 53

Temperature information 54

Temperature information 55

Temperature information 56

Temperature information 57

Temperature information 58

Temperature information 59

Temperature information 60

Temperature information 61

Temperature information 62

Temperature information 63

Temperature information 64

Temperature information 65

Temperature information 66

Temperature information 67

Temperature information 68

Temperature information 69

Temperature information 70

Temperature information 71

Temperature information 72

Temperature information 73

Temperature information 74

Temperature information 75

Temperature information 76

Temperature information 77

Temperature information 78

Temperature information 79

Temperature information 80

Temperature information 81

Temperature information 82

Temperature information 83

Temperature information 84

Temperature information 85

Temperature information 86

Temperature information 87

Temperature information 88

Temperature information 89

Temperature information 90

Temperature information 91

Temperature information 92

Temperature information 93

Temperature information 94

Temperature information 95

Temperature information 96

Temperature information 97

Temperature information 98

Temperature information 99

Temperature information 100

Temperature information 101

Temperature information 102

Temperature information 103

Temperature information 104

Temperature information 105

Temperature information 106

Temperature information 107

Temperature information 108

Temperature information 109

Temperature information 110

Temperature information 111

Temperature information 112

Temperature information 113

Temperature information 114

Temperature information 115

Temperature information 116

Temperature information 117

Temperature information 118

Temperature information 119

Temperature information 120

Temperature information 121

Temperature information 122

Temperature information 123

Temperature information 124

Temperature information 125

Temperature information 126

Temperature information 127

Temperature information 128

Temperature information 129

Temperature information 130

Temperature information 131

Temperature information 132

Temperature information 133

Temperature information 134

Temperature information 135

Temperature information 136

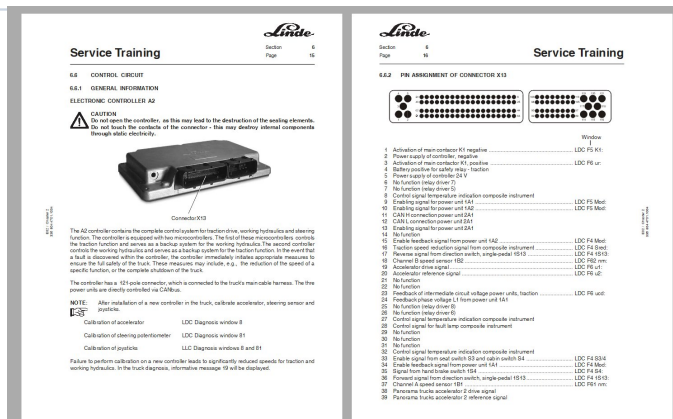
Temperature information 137

Temperature information 138

Temperature information 139

FIELD SERVICE REPORT

Technical Maintenance & Equipment Inspection



0009736330 8:02 AM ✓

solenoid 8:02 AM ✓

0009441242 8:02 AM ✓

valve 8:02 AM ✓

15024404819 8:16 AM ✓

support 8:16 AM ✓

0009264227 x2 8:16 AM ✓

roller 8:16 AM ✓

1874465400 8:17 AM ✓

pully 8:17 AM ✓

9455622215 x2 8:17 AM ✓

9289003301 x2 8:17 AM ✓

15024404807 8:18 AM ✓

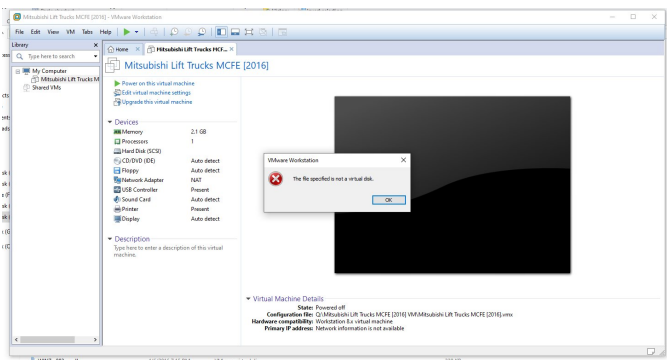
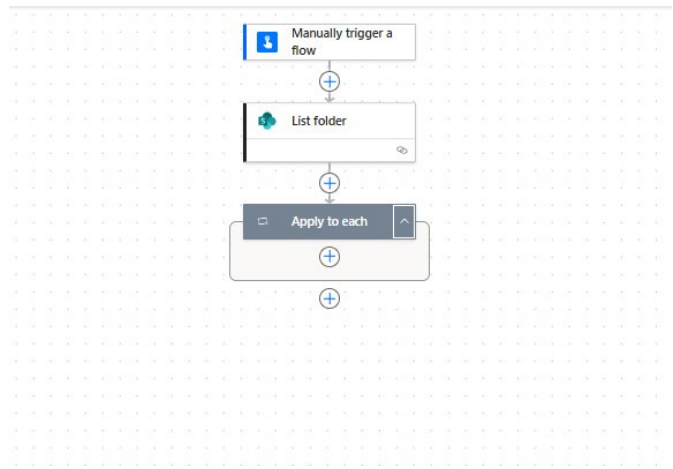
support 8:18 AM ✓

15024405603 8:19 AM ✓

cover 8:19 AM ✓

1874465402 8:19 AM ✓

pully x4 8:19 AM ✓



7. VERIFICATION & SIGN-OFF

Service Engineer: 6685464

Customer Representative: jlhfutkdytsdydjxyrf

Signature: _____ -

Signature: _____ -